

ECED 3201: Introduction to Electronics
Lab 2: Simulating and Characterizing Electronic Circuits

Objectives

Note that some of the diagrams present in this lab were produced with OrCAD's PSpice and will look different when produced in LTSpice. This is not a problem. The objectives of this lab session are:

- Use more advanced simulation options to observe diode properties.
- Simulate and characterize some interesting diode circuits.

1. Extracting Diode Voltage Temperature Dependence

Construct the circuit shown in Fig. 1.

- Q1) Given a constant current, the voltage drop across a silicon diode can be shown to decrease by ~ 2 mV for every 1°C increase in T . Verify this statement using the circuit in Fig.1.

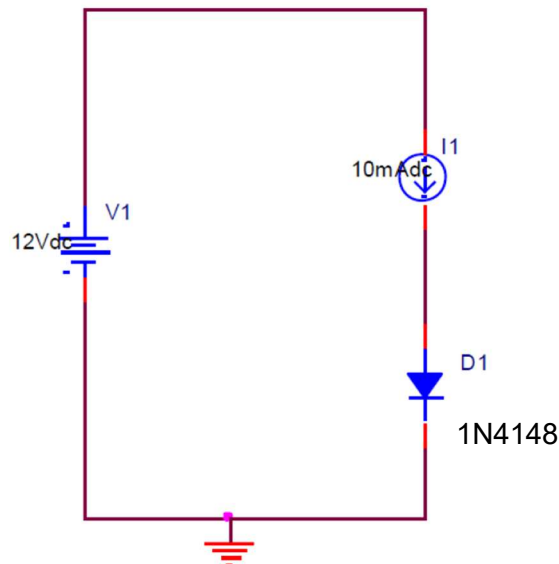


Fig. 1

2. Extracting Diode Voltage Temperature Dependence

Construct the circuit shown in Fig. 2.

- Q2) Use a Transient Simulation to plot the output voltage (V_{out}), when the supply voltage (V_{supply}) can vary by ± 1 V. *Hint: model this variation by adding a sinewave signal with frequency of 1 kHz in series with V_{supply} .*

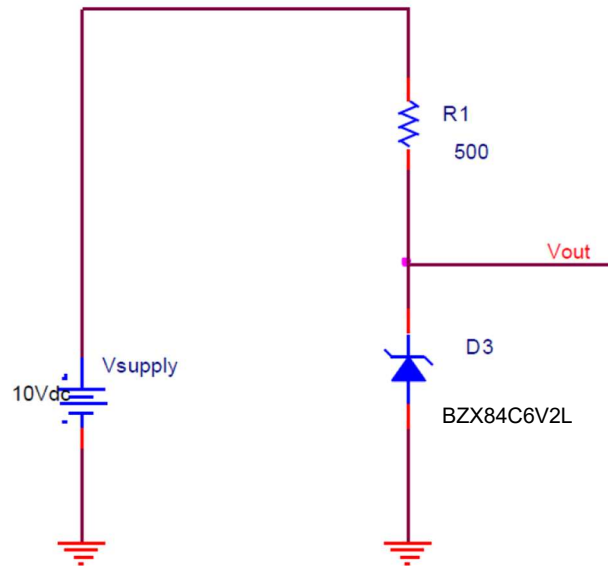


Fig. 2

Q3) Using this transient simulation, find the line regulation of this circuit. Recall: the line regulation is the variation (e.g.: amplitude of AC part) of V_{out} divided by the variation of V_{supply} .

Connect a load R_L to the output of the circuit as shown in Fig. 3.

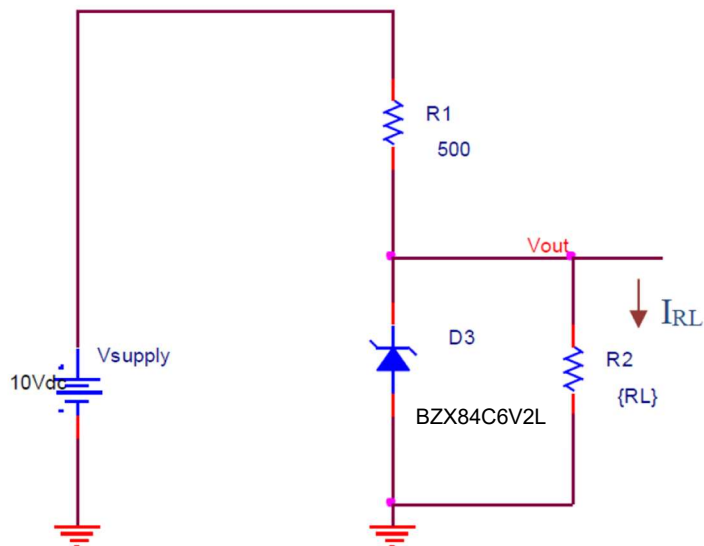


Fig.3

Q4) Use the DC Operating Point (.op) simulation and a parametric sweep (.step param R_L ...) of R_L to plot V_{out} vs. R_L . What is the minimum value of R_L for which the Zener diode remains operating in the Breakdown region? Make a note of the current ($I(R_2)$) at this value of V_{out} and R_L .

- Q5) Change the horizontal axis of your plot from R_L to $I(R_2)$, so that this is a plot of V_{out} vs. I_{RL} . Identify which region of this graph corresponds to the Zener diode remaining in the Breakdown region, then deduce the load regulation of this circuit. Recall: the load regulation is the variation of the output voltage divided by the variation of the output current.

3. Reverse-Biased Diode as a Varactor

Construct the circuit shown in Fig. 4.

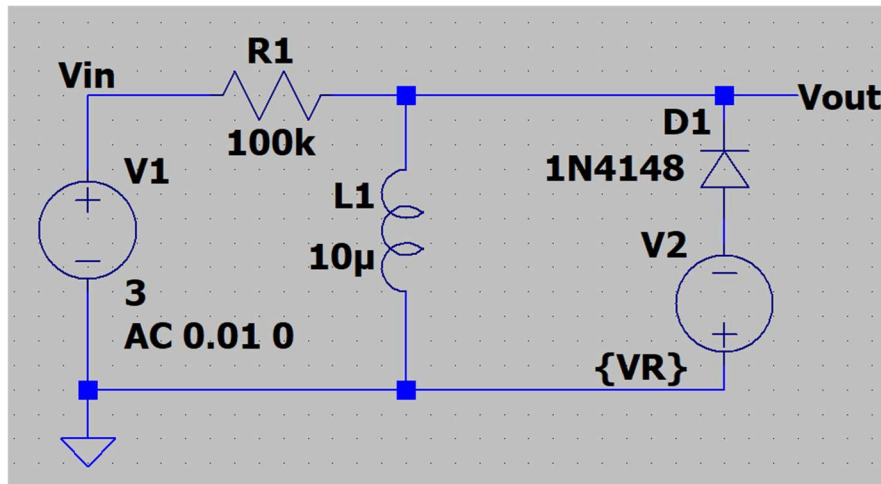


Fig.4

- Q6) Set the simulation to AC Analysis. The voltage source V1 should be configured for a low AC amplitude (0.01), the frequency sweep type should be “Decade”, starting at 10 MHz (10Meg) and ending at 100 MHz with 2000 points per decade. Set a parametric sweep of VR from 0 to 1 V with steps of 0.1V, then plot the transfer function V_{out} / V_{in} vs. frequency.
- Q7) From the obtained results, find the range of equivalent capacitances for this reverse-biased diode. Hint: is there a formula for the frequency of the peak for this type of RLC transfer function?

4. The Super Diode

In lecture notes and the LTSpice tutorial we considered the precision rectifier or super diode. The device acts as a half-rectifier without the voltage drop or temperature dependence of a simple diode rectifier.

Construct the circuit shown in Fig. 5.

- Q8) Using Transient Simulation, plot the output voltage (V_{OUT}) against the input voltage (V_{in}). Is this circuit working as a super diode?

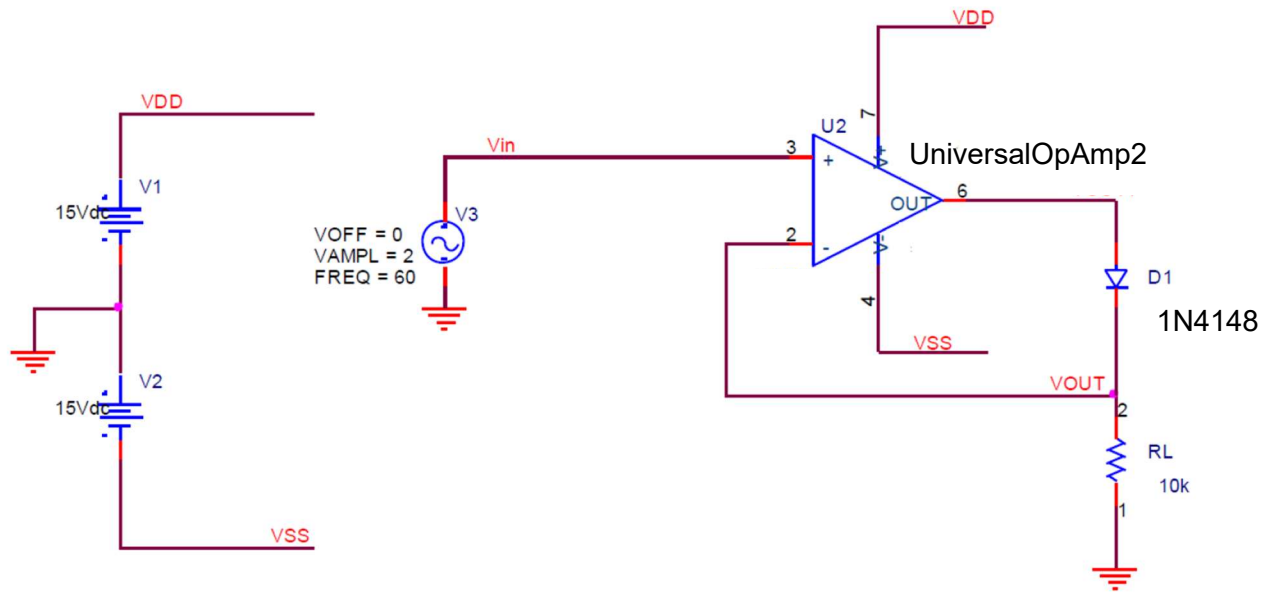


Fig.5

Q9) Reverse the direction of the diode D1. Run the simulation again, but plot the transfer function of the circuit instead, V_{OUT} vs. V_{in} . Describe the output; what is this circuit doing now?

Q10) Perform the following steps carefully:

Return the diode to its original orientation. Duplicate the OpAmp part of the circuit (that is, the OpAmp, the diode, the feedback path to the “-” terminal, and the load resistor) and place the copy below. Route V_{in} so that both amplifiers receive it at their respective “+” terminals. Update the Net Names; the original V_{OUT} should read V_{OUT} , the duplicated V_{OUT} should read V_{OUT2} . Reverse the direction of the duplicated diode as in Q9.

Place a third UniversalOpAmp2 part in front of the original circuits, ensuring it is powered by both V_{DD} and V_{SS} . Wire the output of the original circuit (V_{OUT}) directly to the “+” terminal of the third OpAmp. Remove the ground from the 10 k Ω resistor of the duplicated circuit and connect the resistor to the “-” terminal of the third OpAmp (in other words, V_{OUT2} is connected to “-” through the 10 k Ω resistor). Place an additional 10 k Ω resistor between the output of the third OpAmp and its own “-” terminal. Use a Net Name at the output of the third OpAmp, V_{OUT3} .

Change the simulation type to DC Sweep. Sweep the source V_3 (i.e.: V_{in}) from -1 V to 1 V in steps of 0.1 V. Plot V_{OUT3} vs. V_{in} .

Include a screen capture of your full circuit and the plot of V_{OUT3} vs. V_{in} with your report. What is the function of this circuit?

Lab Report

1. Clearly label all answers (e.g. A5 corresponding to Q5).
2. All graphs must be properly annotated, if you include a graph without labels or units it will not be marked.
3. The report should not contain more than 15 pages.
4. Be succinct and clear, but do not be messy. Answer in complete sentences, make sure margins and formatting are respected throughout the report.
5. Plagiarism will not be tolerated.

Introduction	0.5
Q1	2
Q2	2
Q3	2
Q4	2
Q5	2
Q6	2
Q7	2
Q8	2
Q9	1
Q10	2
Conclusion	0.5
Total	20/20